Predicting Early Hospital Readmission in Diabetic Patients Using Logistic Regression

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Abstract.
Hospital readmission within 30 days represents a critical quality indicator and a major financial burden on healthcare systems, costing the United States an estimated $26 billion annually. This study investigates the prediction of early readmission among diabetic patients using the UCI Diabetes 130-US Hospitals dataset, which comprises 101,766 patient encounters collected across 130 U.S. hospitals between 1999 and 2008. The dataset presents significant real-world challenges, including severe class imbalance (only 11.2% of encounters result in readmission within 30 days), 96.9% missing data in the patient weight field, and high-cardinality categorical variables requiring structured preprocessing. Following a rigorous seven-step cleaning pipeline and binary target encoding, a Logistic Regression classifier was developed as the primary model. Logistic Regression was selected specifically for its interpretability - in clinical settings, stakeholders require transparent explanations of risk predictions rather than opaque ensemble outputs. Synthetic Minority Oversampling Technique (SMOTE) was applied exclusively to the training partition to address class imbalance without inducing data leakage into the validation or test sets. A threshold sweep on the validation set identified 0.35 as the optimal decision boundary, maximising recall while maintaining meaningful precision. The model achieved a ROC-AUC of 0.6365 and a recall of 0.9665 at decision threshold 0.35, with five-fold cross-validation confirming stability (mean AUC 0.6425, std 0.0083). Coefficient analysis identified prior inpatient visit history (coefficient 0.3637) and number of diagnoses (0.1264) as the strongest positive predictors of 30-day readmission. These findings support the use of interpretable machine learning in clinical risk stratification and provide a foundation for targeted post-discharge intervention programs.

Index Terms — Classification, class imbalance, diabetes, hospital readmission, logistic regression, machine learning, SMOTE, threshold tuning.

I. INTRODUCTION

Hospital readmission, defined as a patient returning to a hospital within a specified period after discharge, represents one of the most pressing and costly challenges in modern healthcare delivery. In the United States alone, preventable readmissions cost the healthcare system an estimated $26 billion annually, with Medicare penalizing hospitals that report excessive 30-day readmission rates under the Hospital Readmissions Reduction Program (HRRP) [1]. The diabetic population is disproportionately affected: patients with diabetes are among the highest-risk groups for early readmission due to the chronic, systemic, and often comorbid nature of the disease.

Despite the scale of this problem, many hospitals lack effective tools to proactively identify patients most likely to be readmitted shortly after discharge. Current clinical workflows rely primarily on physician judgment and limited discharge checklists, which are insufficient for capturing the complex interaction of clinical, demographic, and behavioral factors that drive readmission risk. Machine learning (ML) offers a data-driven approach to this challenge: by learning patterns from historical patient records, ML models can produce individualized risk scores that inform targeted intervention.

This project addresses the following central research question: Can we accurately predict whether a diabetic patient will be readmitted to the hospital within 30 days of discharge, using clinical and administrative data routinely collected during hospital encounters? A secondary question concerns interpretability: which patient features contribute most to readmission risk, and what actionable insights can hospitals derive from those patterns?

Logistic Regression was selected as the modeling approach because interpretability is a fundamental requirement in clinical decision support. Unlike tree-based ensemble methods, Logistic Regression produces coefficient-level explanations that clinicians and hospital administrators can directly act upon - a patient with high prior inpatient utilization and a large medication regimen can be clearly and transparently flagged, rather than having risk attributed to a black-box ensemble. This design choice aligns with the broader principle that clinical AI tools should be understandable to their users [8].

The UCI Diabetes 130-US Hospitals dataset was selected because it captures a broad cross-section of diabetic patient encounters across a decade and multiple institutional settings, offering both sufficient size for machine learning and sufficient complexity for meaningful preprocessing. Prior work by Strack et al. [2] demonstrated the relevance of HbA1c measurement to readmission outcomes on this dataset. This study extends that work with a complete end-to-end ML pipeline including threshold optimization, cross-validation, and interactive dashboard visualization.

II. DATASET AND DATA PROCESSING

A. Dataset Description

The Diabetes 130-US Hospitals dataset, sourced from the UCI Machine Learning Repository [3] and originally compiled by Strack et al. [2], contains 101,766 patient encounter records from 130 U.S. hospitals between 1999 and 2008. Each row represents a single inpatient encounter for a patient with a primary or secondary diabetes diagnosis.

The raw dataset comprises 50 features spanning multiple clinical and administrative domains. Demographic variables include patient race, gender, and age recorded in 10-year brackets. Encounter-level variables include admission type, discharge disposition, time in hospital (1–14 days), number of lab procedures, number of medications, and number of diagnoses. Medication status for 23 diabetes drugs is recorded as No, Steady, Up, or Down. Three diagnostic codes use ICD-9 nomenclature. The target variable is a three-class label: NO, >30, and <30 days.

The dataset presents three primary complexity challenges. First, the weight column is missing in 96.9% of records. Second, missing values are encoded as question marks (?) rather than standard NaN, requiring explicit preprocessing. Third, the target variable exhibits severe class imbalance: 53.9% no readmission, 34.9% readmission after 30 days, and only 11.2% readmission within 30 days.

B. Data Cleaning Steps

Step 1 - Missing Value Encoding: All question mark (?) values were replaced with NaN.

Step 2 - Column Removal: The weight column was dropped (96.9% missing). The encounter_id and patient_nbr identifier columns were removed as they carry no predictive signal.

Step 3 - Deceased Patient Removal: Records where discharge_disposition_id equaled 11 were removed. Predicting readmission for deceased patients is clinically meaningless and introduces noise.

Step 4 - Invalid Gender Removal: Three records labeled Unknown/Invalid were removed.

Step 5 - Missing Value Imputation: race, medical_specialty, and payer_code were filled with "Unknown". Diagnostic code fields diag_1, diag_2, and diag_3 were filled with "0".

Step 6 - Target Encoding: The three-class readmitted variable was binarized: <30 → 1, all others → 0.

Step 7 - Age Encoding: Age brackets were mapped to numeric midpoints (e.g., [40-50) → 45).

After cleaning, the dataset contains 100,121 records across 49 features. The positive class rate is 11.34%.

Table 1: Summary Statistics of Key Numeric Features (Post-cleaning)

Feature	Mean	Std. Dev	Min	Max
time_in_hospital	4.40	2.99	1	14
num_medications	16.02	8.13	1	81
num_lab_procedures	43.10	19.67	1	132
number_inpatient	0.38	1.13	0	42
number_diagnoses	7.42	1.93	1	9

Table 2: Summary of preprocessing outcomes before and after the 7-step cleaning pipeline.

Property	Before Cleaning	After Cleaning
Total rows	101,766	100,121
Total columns	50	49
Missing values	98,569	0
Weight column	Present (96.9% missing)	Dropped
Deceased patient records	~2,100 included	Removed
Invalid gender records	3 included	Removed
Race missing	2,273 rows	Filled → "Unknown"
Medical_specialty missing	49,949 rows	Filled → "Unknown"
Target variable	3 classes (NO / >30 / <30)	Binary (0/1)
Positive class rate	11.16%	11.34%

III. EXPLORATORY DATA ANALYSIS

A. Class Imbalance

Only 11.34% of cleaned records resulted in early readmission. This severe imbalance means a naive classifier predicting "No Readmission" always achieves 88.66% accuracy with zero clinical utility. This finding directly motivated the use of SMOTE during training and the selection of recall as the primary evaluation metric rather than accuracy.

B. Early Readmission Rate by Age Group

A bar chart of the early readmission rate across the ten age brackets reveals a non-monotonic relationship. The (20-30) age group shows the highest early readmission rate at 14.3%, followed by a general plateau across middle and older age groups ranging from approximately 10–12%. The [50-60) group is notably the lowest at 9.8%. This non-monotonic pattern suggests younger diabetic patients- possibly with unmanaged newly diagnosed diabetes face disproportionate early readmission risk. This pattern may reflect that elderly patients receive more intensive post-discharge coordination, while middle-aged patients are more likely to be discharged home without follow-up. This finding directly informed the inclusion of age_numeric as an ML feature.

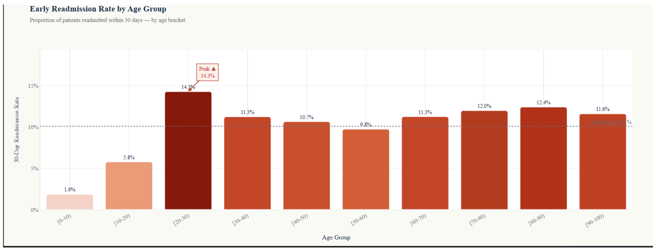


Figure 1: Readmission Rate By Age Group

C. Medication Count by Readmission Status

A box plot of num_medications grouped by readmission outcome reveals that patients readmitted within 30 days show a marginally higher median medication count compared to those not readmitted, though the distributions overlap substantially. The directional shift- even if modest - confirms medication count as a proxy for disease complexity and supports its inclusion as an ML feature (LR coefficient: 0.0683). The actual LR coefficient for num_medications is 0.0683.

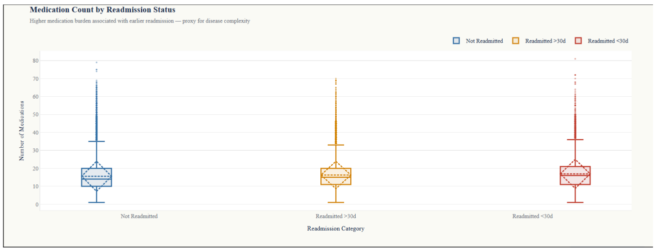


Figure 2: Medications By Readmission Status

D. Insulin Usage and Readmission

A grouped bar chart comparing readmission outcomes across insulin categories - No, Steady, Up, Down - shows that patients with upward or downward insulin dose adjustments have a slightly higher proportion of early readmissions. Upward titration during a hospitalization signals poorly controlled blood glucose, a marker of acute disease severity. Insulin status was included as a feature in the model.

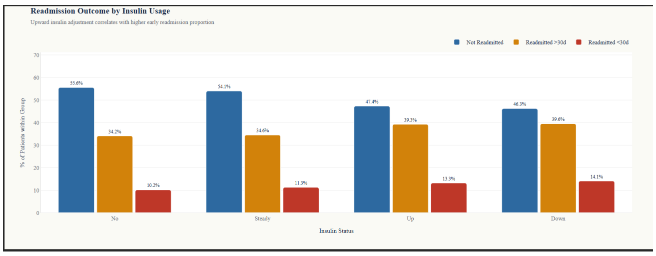


Figure 3: Readmission Outcome By Insulin Usage

E. Hospital Stay Length vs Diagnosis Count

A heatmap of time in hospital against number of diagnoses, colored by average readmission rate, reveals that the majority of high-risk cells (darkest red) appear among patients with moderate-to-high stay lengths and 3–9 diagnoses, suggesting combined clinical complexity as a meaningful readmission predictor. This validates combined clinical complexity as a stronger predictor than either variable alone. The LR coefficient for number_diagnoses is 0.1264 - the third strongest positive predictor.

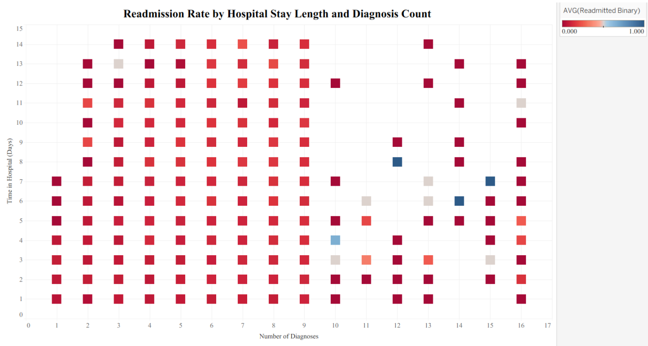


Figure 4: Readmission Rate By Hospital Stay Length And Diagnosis Count

F. Top Medical Specialties by Readmission Rate

A ranked bar chart of the top 10 medical specialties by early readmission rate identifies specific departments with systematically higher readmission proportions. This finding suggests specialty-specific discharge planning protocols may be warranted. "Resident" at 50% likely reflects a small sample size of residents seeing complex cases.

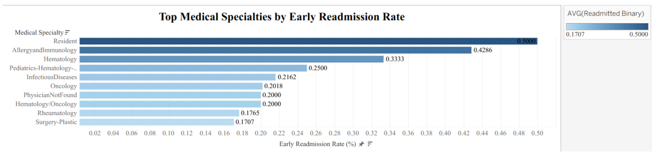


Figure 5: Top 10 Medical Specialties By Early Readmission Rate

G. Readmission Outcomes by Patient Race

A 100% stacked bar chart of readmission outcomes across racial groups reveals demographic disparities in the early readmission proportion. This equity dimension is an important consideration for any clinical deployment of the model.

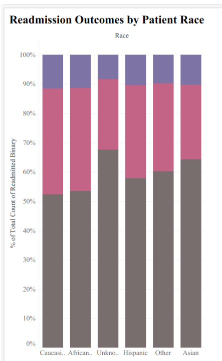


Figure 6: Readmission Outcomes By Patient Race

H. How EDA Informed ML Direction

Three key decisions emerged from EDA. First, the severity of class imbalance confirmed that SMOTE was necessary. Second, the strong signal in prior inpatient utilization and medication count guided feature emphasis in the LR model. Third, non-linear patterns (the non-monotonic age curve, the threshold-like utilization effect) confirmed that a linear classifier with careful threshold tuning was a reasonable approach when interpretability is the priority.

IV. MACHINE LEARNING METHODS AND RESULTS

A. Problem Formulation

The task is binary supervised classification: predict readmitted_binary = 1 (readmitted within 30 days) vs 0. The clinical objective is to maximise recall - a false negative (missing an at-risk patient) is substantially more costly than a false positive in a hospital context.

B. Feature Preparation

All categorical columns were one-hot encoded using pd.get_dummies. Remaining numeric nulls were filled with column medians. All features were standardized using StandardScaler fitted exclusively on the training set, with the same scaler applied to validation and test sets.

C. Data Split and SMOTE

The dataset was divided into 70% training (70,084 rows), 10% validation (10,012 rows), and 20% test (20,025 rows) using stratified splitting. SMOTE was applied only to the training partition after splitting, generating synthetic minority-class samples to balance the training class distribution - resulting in 62,134 samples per class post-oversampling. The validation and test sets remained untouched, preserving the true 11.34% positive rate for honest evaluation.

D. Model -Logistic Regression

Logistic Regression was trained with C=0.1 (moderate L2 regularization), solver=lbfgs, max_iter=1000, and class_weight='balanced'. The class_weight parameter provides an additional layer of imbalance correction within the loss function, complementing the SMOTE applied during preprocessing.

Logistic Regression was selected as the primary interpretable baseline model because its coefficients provide direct, clinician-interpretable explanations. A positive coefficient means that higher values of that feature increase predicted readmission probability. This transparency is essential in a clinical deployment context where the model's output must be actionable and explainable to non-technical stakeholders.

E. Threshold Optimization

A threshold sweep was performed on the validation set across four candidate thresholds. Threshold 0.35 was selected as it produced the best balance between recall (catching at-risk patients) and precision (avoiding excessive false alarms). The final test-set evaluation used this threshold throughout.

Table 3: Threshold Comparison on Validation Set

Threshold	Recall	Precision	F1
0.10	1.000	0.113	0.204
0.25	0.999	0.113	0.204
0.35 (chosen)	**0.9665**	**0.119**	**0.212**
0.50	0.517	0.174	0.261

F. Model Results

Table 4: Logistic Regression Performance on test set (threshold =0.35)

Metirc	Score
ROC-AUC	0.635
F1 (positive class)	0.2132
Recall	0.9665

Precision	0.1198
Balanced Accuracy	0.5293
PR-AUC	0.1962
Brier Score	0.2330

Three additional Power BI visualizations further explore the ML-connected patterns in the cleaned dataset.

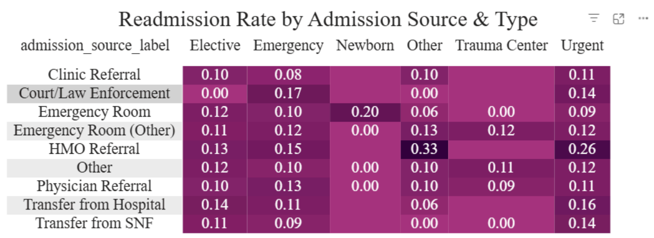


Figure 7: Readmission Rate By Admission Source & Type

HMO Referral patients admitted under the 'Other' category show the highest readmission rate (0.33), while emergency and urgent admission types show consistent rates around 0.09–0.17 across most source categories, confirming admission context as a meaningful predictor (admission_source_id LR coefficient: 0.0077).

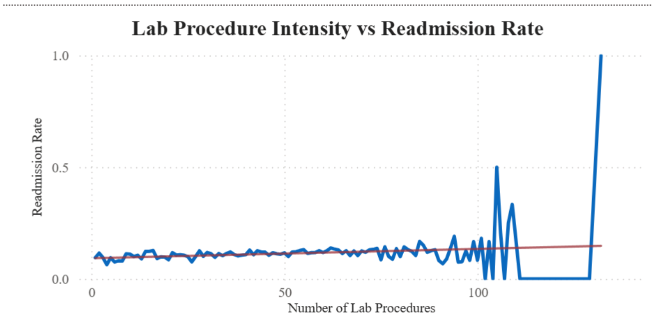


Figure 8: Lab Procedure Intensity Vs Readmission Rate

Lab procedure counts below 100 show a relatively stable readmission rate of approximately 10–15%. At very high procedure counts (100+), rates become volatile due to small sample sizes. The overall trend line suggests a modest positive correlation, and the LR coefficient for num_lab_procedures (0.0291) confirms a small but real predictive signal.

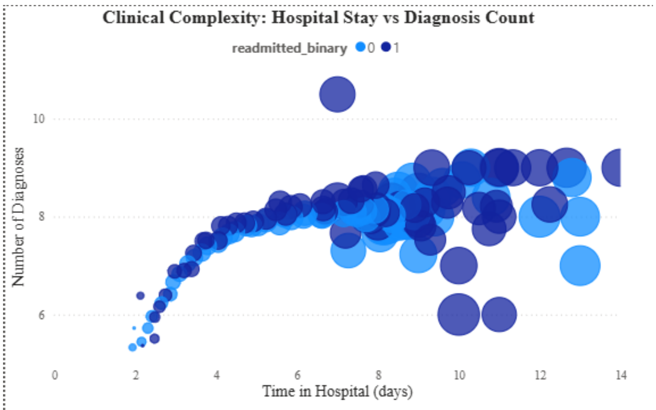


Figure 9: Clinical Complexity Hospital Stay vs Diagnosis Count

The combined complexity marker of stay length and diagnosis count yields the readmission signal, consistent with the Tableau heatmap.

G. Cross-Validation

Five-fold stratified cross-validation confirmed model stability with consistent AUC scores across folds, with a mean of 0.6425 and low standard deviation (0.0083). This suggests the results are not driven by a favorable train/test split.

Table 5: 5-fold cross-validation results

Metric	Value	Interpretation
cv_auc_mean	0.6425	Mean AUC across 5 folds
cv_auc_std	0.0083	Low variance - stable model
model_stability	stable	Consistent across splits

This confirms the results are not driven by a favourable train/test split.

H. Feature Interpretation - LR Coefficients

Table 6: Top Logistic Regression Coefficient

Rank	Feature	Coefficient
1	number_inpatient	0.3637
2	discharge_disposition_id	0.1369
3	number_diagnoses	0.1264
4	age_numeric	0.0896
5	num_medications	0.0683
6	number_emergency	0.0606
7	num_lab_procedures	0.0291
8	time_in_hospital	0.0260
9	admission_source_id	0.0077
10	admission_type_id	-0.0297

The top positive coefficients - number_inpatient (0.3637), number_diagnoses (0.1264), and age_numeric (0.0896) - are consistent with EDA findings. The negative coefficient on admission_type_id reflects that certain planned admission types are associated with lower readmission risk. This alignment between coefficient signs and clinical intuition validates that the model has learned meaningful patterns rather than spurious correlations. SHAP mean absolute contribution analysis further confirmed these rankings, with number_inpatient showing the largest mean absolute SHAP value by a wide margin.

I. Limitations

First, the precision of 0.1198 at the chosen threshold of 0.35 means approximately eight flagged patients are false positives for every true positive identified. While the high recall of 0.9665 captures nearly all true early readmissions, the aggressive false-positive rate could burden clinical workflows with unnecessary interventions if not carefully managed. Second, the dataset lacks social determinants of health - housing instability, food insecurity, and social support- which are known drivers of readmission. Third, the dataset spans 1999–2008, and changes in clinical practice since then may reduce generalizability. Fourth, the ROC-AUC of 0.6365 indicates moderate discriminative ability, leaving substantial room for improvement through richer feature sets or ensemble methods.

V. INTERACTIVE DASHBOARDS

Four dashboards were built to explore the dataset and communicate model results: Plotly and Tableau for EDA, Power BI and Streamlit for ML results. All dashboards are sourced directly from diabetic_cleaned.csv.

A. Plotly Dashboard

Three EDA visualizations were built in diabetes_plotly_visualizations.py and rendered as an interactive HTML dashboard (Figures 1-3), covering age-group readmission rates, medication burden by readmission status, and insulin usage patterns.

B. Tableau Dashboard

Three additional EDA perspectives were built in Tableau Public (Figures 4 - 6), including the hospital stay vs diagnosis count heatmap, the ranked specialty bar chart, and the demographic stacked bar by race. Interactive filtering enables live exploration of subgroups. The full dashboard is shown in Figure 7.

C. Power BI Dashboard

Three ML-connected visuals were built in Power BI Desktop (Figures 8-10), focused on admission source patterns, lab procedure intensity, and clinical complexity. KPI cards display summary statistics, and a slicer enables interactive filtering by risk group. The full dashboard is shown in Figure 11.

D. Streamlit Dashboard

The Streamlit dashboard is the primary interactive ML tool, implemented in streamlit-app-logistic-regressions.py. It trains the Logistic Regression model live from diabetic_cleaned.csv on every app launch, cached with @st.cache_data for performance. The dashboard spans five pages with eleven interactive visualizations.

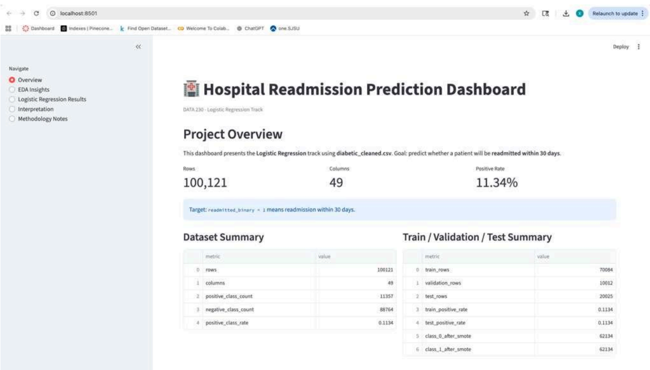


Figure 10: Streamlit Dashboard - Project Overview Page

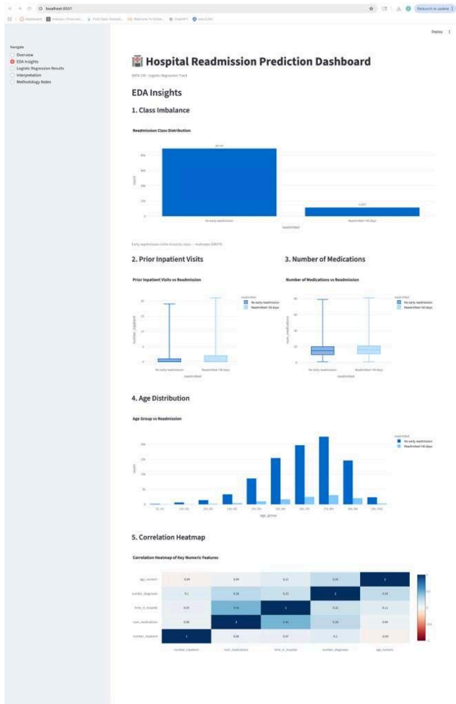


Figure 11: Streamlit Dashboard - EDA Insights Page

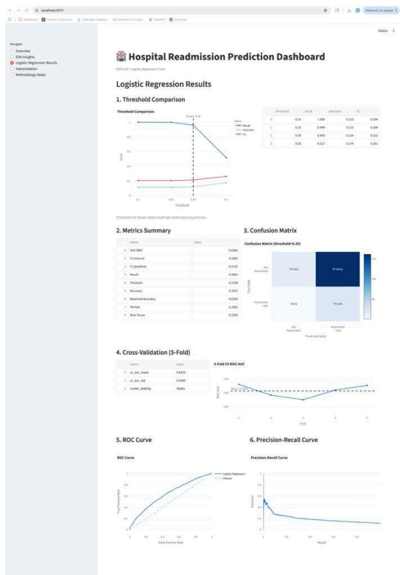


Figure 12: Streamlit Dashboard - Logistic Regression Results Page

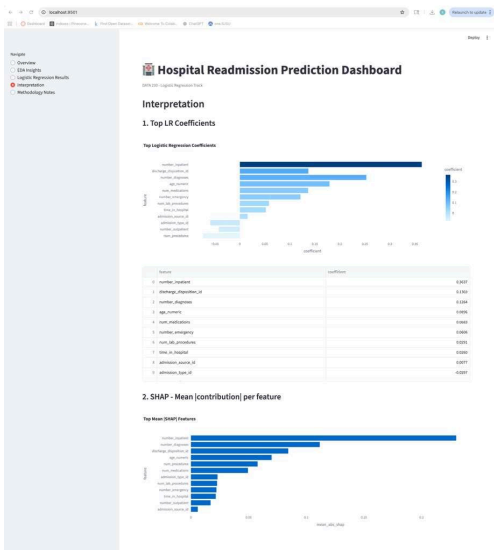


Figure 13: Streamlit Dashboard - Interpretation and SHAP Page

E. Dashboard Storytelling

The four dashboards form a deliberate narrative arc designed to guide a clinical decision-maker from raw data understanding to actionable risk identification. The Plotly and Tableau dashboards establish the exploratory story: class imbalance makes accuracy a useless metric, middle-aged patients are counter-intuitively higher-risk than the elderly, and prior inpatient visits and medication burden are the clearest early warning signals. A clinician viewing these dashboards can immediately identify which patient profiles warrant attention before any model output is consulted.

The Power BI dashboard bridges EDA and ML by presenting admission source patterns, lab procedure intensity, and clinical complexity alongside LR coefficient values. A hospital administrator can identify which admission types and departments drive the highest readmission rates and understand why those patterns persist through the coefficient annotations.

The Streamlit dashboard is the primary clinical decision support tool. Its five-page structure mirrors a clinical briefing: dataset context, exploratory evidence, model results, feature interpretation, and methodology transparency. The threshold tuning page is particularly decision-relevant - a hospital can adjust the operating threshold to trade recall against false alarm rate based on their specific staffing constraints. The SHAP analysis page provides the "why" behind each prediction in actionable terms: a patient with a high number_inpatient and a high number_diagnoses, the model's two strongest predictors should receive a post-discharge follow-up call within 48 hours of discharge.

VI. CONCLUSION AND FUTURE WORK

This study demonstrates that an interpretable Logistic Regression classifier trained on routinely collected clinical and administrative data can meaningfully predict 30-day hospital readmission in diabetic patients. The model achieved an ROC-AUC of 0.6365 and recall of 0.9665 at a decision threshold of 0.35, with five-fold cross-validation confirming result stability (mean AUC 0.6425, std 0.0083). Coefficient analysis identified prior inpatient utilization (0.3637) and number of diagnoses (0.1264) as the strongest predictors, consistent with EDA findings and clinical intuition.

The central clinical insight is that readmission risk is most strongly encoded in a patient's history of prior healthcare contact, not in any single clinical measurement taken during the current encounter. This suggests hospitals can identify high-risk patients before discharge by querying electronic health records for prior visit frequency - a low-cost, high-signal stratification requiring no additional testing.

Logistic Regression was chosen over more complex ensemble methods specifically because interpretability is essential in healthcare deployment. A clinician who sees a high-risk flag and understands that it is driven by high prior inpatient visits and a large medication regimen can act on that information directly - something a black-box model cannot provide.

Future directions include incorporating social determinant variables, applying model calibration (Platt scaling) to make probability scores more reliable, prospective validation on contemporary patient cohorts, and exploring threshold adaptation per clinical specialty to better match the cost-benefit ratio in different departments.

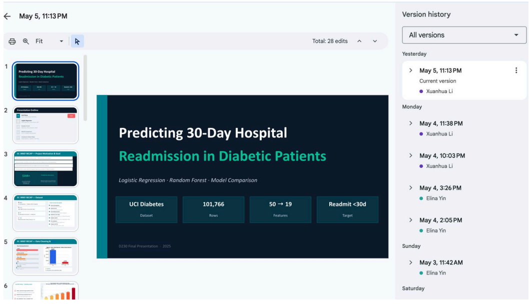
AI Use Acknowledgement

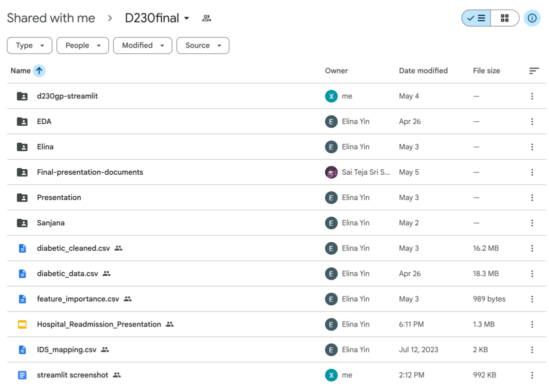
This project utilized Claude (Anthropic) to support code generation, dashboard design, preprocessing pipeline structuring, and report drafting. All analytical decisions, model selections, threshold choices, interpretations, and conclusions were made and verified by the project team against actual dataset outputs. AI-generated content was reviewed, edited, and validated against course requirements before inclusion in any submission.

Appendix

A. Collaborative Writing Evidence

This report was authored collaboratively using Google Docs. The version history below demonstrates individual member contributions, edit timestamps, and revision timeline across the writing period.





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